Statistical Concepts, Linear Algebra & Machine Learning

Study Notes

Part 1: Statistical Concepts

Measures of Central Tendency

Mean (Average)

- **Definition:** Sum of all values divided by the number of values
- **Formula:** $\mu = (\Sigma x) / n$, where n is the number of observations
- **Use:** Most common measure of central tendency
- **Limitation:** Sensitive to outliers

Median

- **Definition:** The middle value when data is arranged in order
- **How to find:** For odd n: the middle value; For even n: average of two middle values
- **Advantage:** Not affected by extreme outliers
- **Use:** Preferred for skewed distributions

Mode

- **Definition:** The value that appears most frequently in a dataset
- **Types:** Unimodal (one mode), Bimodal (two modes), Multimodal (more than two modes)
- **Use:** Best for categorical data
- **Note:** A dataset may have no mode or multiple modes

Outliers

- **Definition:** Data points that differ significantly from other observations
- **Impact:** Can distort mean and other statistical measures
- **Detection methods:** Values beyond 1.5 × IQR, Z-score method ($|z| > 3$), Visual inspection using box plots
- **Handling:** May be removed, transformed, or analyzed separately

Measures of Spread

Range

- **Definition:** Difference between maximum and minimum values
- **Formula:** Range = Max - Min
- **Limitation:** Only considers two extreme values, ignores distribution

Average Deviation

- **Definition:** Average of the absolute differences from the mean
- **Formula:** $AD = \Sigma|x - \mu| / n$
- **Use:** Simple measure of variability

Absolute Deviation

- **Definition:** The absolute difference of each data point from the mean
- **Formula:** $|x - \mu|$
- **Purpose:** Removes negative signs to measure distance from center

Squared Deviation

- **Definition:** The square of the difference between each value and the mean
- **Formula:** $(x - \mu)^2$
- **Purpose:** Foundation for variance and standard deviation
- **Advantage:** Penalizes larger deviations more heavily

Standard Deviation

- **Definition:** Square root of the average of squared deviations
- **Formula:**

- Population: $\sigma = \sqrt{}[\Sigma(x - \mu)^2 / N]$

- Sample: $s = \sqrt{}[\Sigma(x - x■)^2 / (n-1)]$

- **Interpretation:** Measures how spread out data is from the mean
- **Properties:** Always non-negative; Higher value = more spread; In normal distribution: ~68% data within $\pm 1\sigma$, ~95% within $\pm 2\sigma$

Probability Theory

Basic Concepts

- **Probability:** Measure of likelihood of an event occurring
- **Range:** $0 \leq P(A) \leq 1$
- **Formula:** P(A) = (Number of favorable outcomes) / (Total number of outcomes)

Key Rules

- **Addition Rule:** P(A or B) = P(A) + P(B) - P(A and B)
- **Multiplication Rule:** P(A and B) = P(A) × P(B|A)
- **Complement Rule:** P(A') = 1 - P(A)

Important Concepts

- **Independent Events:** P(A and B) = P(A) × P(B)
- **Conditional Probability:** P(A|B) = P(A and B) / P(B)
- **Random Variable:** Variable whose value depends on outcomes of a random phenomenon
- **Expected Value:** E(X) = Σ[x × P(x)]

Part 2: Linear Algebra Review

Vectors

Definition

- **Vector:** Ordered array of numbers (can be row or column)
- **Notation:** $\mathbf{v} = [v_1, v_2, ..., v_n]$
- **Dimension:** Number of components in the vector

Vector Operations

- **Addition:** $\mathbf{u} + \mathbf{v} = [u_1+v_1, u_2+v_2, ..., u_n+v_n]$
- **Scalar Multiplication:** $c\mathbf{v} = [cv_1, cv_2, ..., cv_n]$
- **Dot Product:** $\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + ... + u_nv_n$
- **Magnitude:** $||\mathbf{v}|| = \sqrt{}(v_1^2 + v_2^2 + ... + v_n^2)$

Matrices

Definition

- **Matrix:** Rectangular array of numbers arranged in rows and columns
- **Notation:** $A = [a_{ij}]$, where i = row index, j = column index
- **Dimensions:** $m \times n$ (m rows, n columns)

Matrix Operations

Addition and Subtraction

- Only possible when matrices have same dimensions
- Element-wise operation: $C = A + B$ where $c_{ij} = a_{ij} + b_{ij}$

Scalar Multiplication

- Multiply each element by scalar: $kA = [k \cdot a_{ij}]$

Matrix Multiplication

- $A(m \times n) \times B(n \times p) = C(m \times p)$
- Element formula: $c_{ij} = \Sigma(a_{ik} \times b_{kj})$
- **Not commutative:** $AB \neq BA$ (in general)

Hadamard Product (Element-wise)

- A ■ $B = [a_{ij} \times b_{ij}]$
- Requires same dimensions

Matrix Properties

Important Properties

- **Associative:** $(AB)C = A(BC)$
- **Distributive:** $A(B + C) = AB + AC$
- **Identity Matrix (I):** $AI = IA = A$

- Square matrix with 1s on diagonal, 0s elsewhere

- **Zero Matrix:** $A + 0 = A$

Non-Commutative

- $AB \neq BA$ (generally)
- Order matters in matrix multiplication

Matrix Inverse

Definition

- **Inverse (A^{-1}):** Matrix such that $AA^{-1} = A^{-1}A = I$
- Only exists for square, non-singular matrices

Properties

- $(A^{-1})^{-1} = A$
- $(AB)^{-1} = B^{-1}A^{-1}$
- $(A^T)^{-1} = (A^{-1})^T$

Uses

- Solving systems of linear equations: $Ax = b \rightarrow x = A^{-1}b$
- Essential in many ML algorithms

Matrix Transpose

Definition

- **Transpose (A^T):** Flip matrix over its diagonal
- Element formula: $(A^T)_{ij} = A_{ji}$
- Dimensions: If A is m×n, then A^T is n×m

Properties

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$

- $(AB)^T = B^T A^T$
- $(kA)^T = kA^T$

Special Matrices

- **Symmetric:** $A = A^T$
- **Orthogonal:** $AA^T = A^T A = I$

Part 3: Introduction to Machine Learning

What is Machine Learning?

Definition

Machine Learning is a subset of Artificial Intelligence that enables systems to learn and improve from experience without being explicitly programmed. It focuses on developing algorithms that can access data and use it to learn for themselves.

Key Characteristics

- **Data-driven:** Learns patterns from data
- **Adaptive:** Improves performance with more data
- **Automated:** Discovers patterns without explicit programming
- **Predictive:** Makes predictions or decisions based on learned patterns

Applications

- Image and speech recognition
- Recommendation systems
- Fraud detection
- Medical diagnosis
- Autonomous vehicles
- Natural language processing

Three Main Approaches to Machine Learning

1. Supervised Learning

Definition

Learning from labeled data where each training example has an input-output pair. The algorithm learns to map inputs to outputs.

How it Works

1. Training data contains both features (X) and labels (y)
2. Algorithm learns the mapping function: $f(X) \rightarrow y$
3. Model is evaluated on unseen test data
4. Goal: Minimize prediction error

Types of Problems

Classification

- **Goal:** Predict categorical labels
- **Examples:** Email spam detection (spam/not spam), Disease diagnosis (positive/negative), Image classification (cat/dog/bird)
- **Algorithms:** Logistic Regression, Decision Trees, SVM, Neural Networks

Regression

- **Goal:** Predict continuous numerical values
- **Examples:** House price prediction, Stock price forecasting, Temperature estimation
- **Algorithms:** Linear Regression, Polynomial Regression, Neural Networks

Key Characteristics

- Requires labeled training data
- Teacher/supervisor provides correct answers
- Performance measured by accuracy on unseen data
- Can be computationally expensive to label data

2. Unsupervised Learning

Definition

Learning from unlabeled data where the algorithm discovers hidden patterns or structures without guidance.

How it Works

1. Training data contains only features (X), no labels
2. Algorithm identifies patterns, groupings, or structures
3. No predefined correct answer
4. Goal: Discover inherent structure in data

Types of Problems

Clustering

- **Goal:** Group similar data points together
- **Examples:** Customer segmentation, Document categorization, Image compression
- **Algorithms:** K-Means, Hierarchical Clustering, DBSCAN

Dimensionality Reduction

- **Goal:** Reduce number of features while preserving information

- **Examples:** Data visualization, Feature extraction, Noise reduction
- **Algorithms:** PCA, t-SNE, Autoencoders

Association

- **Goal:** Find rules describing relationships in data
- **Examples:** Market basket analysis, Recommendation systems
- **Algorithms:** Apriori, FP-Growth

Key Characteristics

- No labeled data required
- Exploratory in nature
- Discovers hidden patterns
- Often used for data preprocessing

3. Reinforcement Learning

Definition

Learning through interaction with an environment, where an agent learns to make decisions by receiving rewards or penalties for its actions.

How it Works

1. Agent observes current state of environment
2. Agent takes an action
3. Environment provides feedback (reward/penalty)
4. Agent updates its strategy to maximize cumulative reward
5. Process repeats iteratively

Key Components

- **Agent:** The learner/decision maker
- **Environment:** The world the agent interacts with
- **State:** Current situation of the agent
- **Action:** Choices available to the agent
- **Reward:** Feedback signal (positive or negative)
- **Policy:** Strategy for selecting actions

Examples

- Game playing (Chess, Go, video games)
- Robotics and autonomous navigation

- Resource management
- Traffic signal control
- Personalized recommendations

Key Characteristics

- Learning through trial and error
- Delayed rewards (actions now affect future)
- Balances exploration vs exploitation
- No explicit training data, learns from experience
- Goal: Maximize long-term cumulative reward

Comparison of the Three Approaches

Aspect	Supervised	Unsupervised	Reinforcement
Data	Labeled	Unlabeled	Interaction-based
Goal	Predict output	Discover patterns	Maximize reward
Feedback	Correct answers	None	Reward/penalty
Example	Email classification	Customer grouping	Game playing
Use case	Known outcomes	Exploration	Sequential decisions

Summary

Statistical Concepts provide the foundation for understanding data distribution, variability, and probability—essential for analyzing ML algorithm performance.

Linear Algebra forms the mathematical backbone of ML, enabling efficient computation with large datasets through vector and matrix operations.

Machine Learning offers three distinct learning paradigms:

- **Supervised:** Learn from examples with answers
- **Unsupervised:** Discover hidden patterns
- **Reinforcement:** Learn through experience and feedback

Each approach suits different types of problems and data availability scenarios.